ETH zürich

University of Zurich[UZH]

A meritocratic network formation model for the rise of social media influencers

Nicolò Pagan

Post-Doctoral Researcher, ETH - University of Zürich

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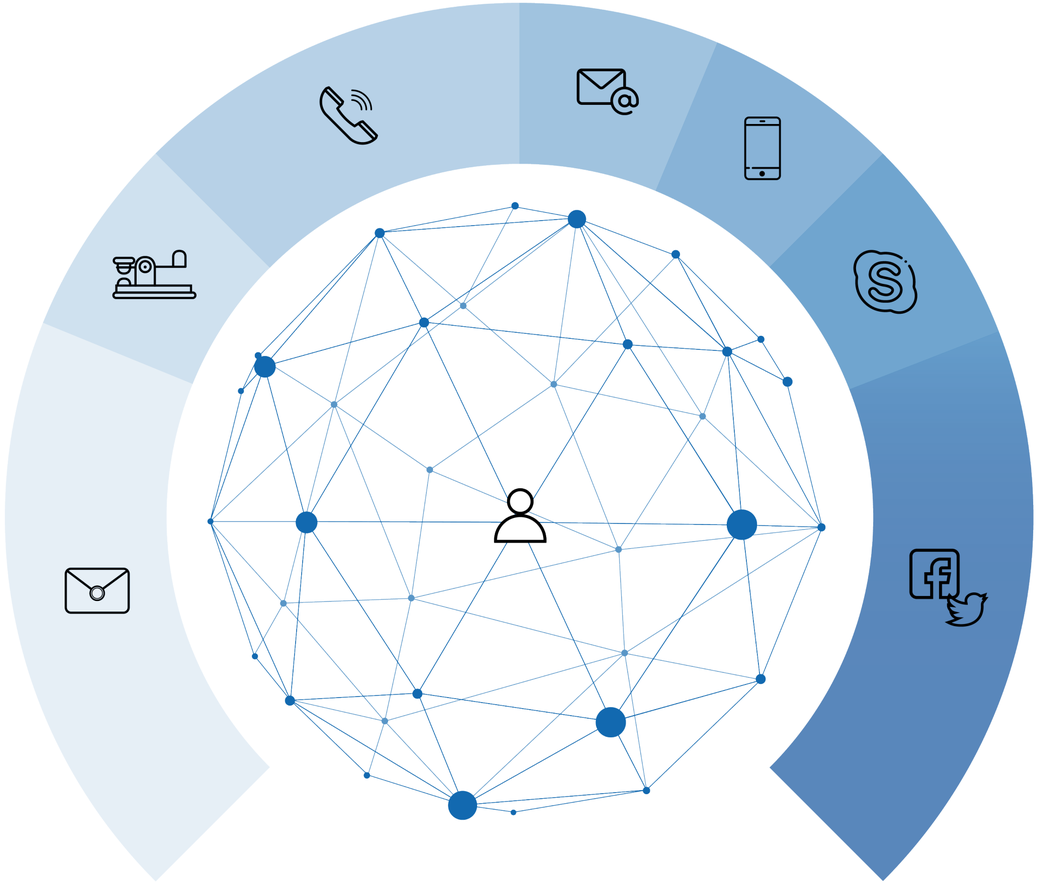
Social Networks in Human History

Man is by nature a social animal; an individual who is unsocial naturally and not accidentally is either beneath our notice or more than human. Society is something that precedes the individual. Anyone who either cannot lead the common life or is so self-sufficient as not to need to, and therefore does not partake of society, is either a beast or a god.

Aristotle, Politics

Social Networks in Human History

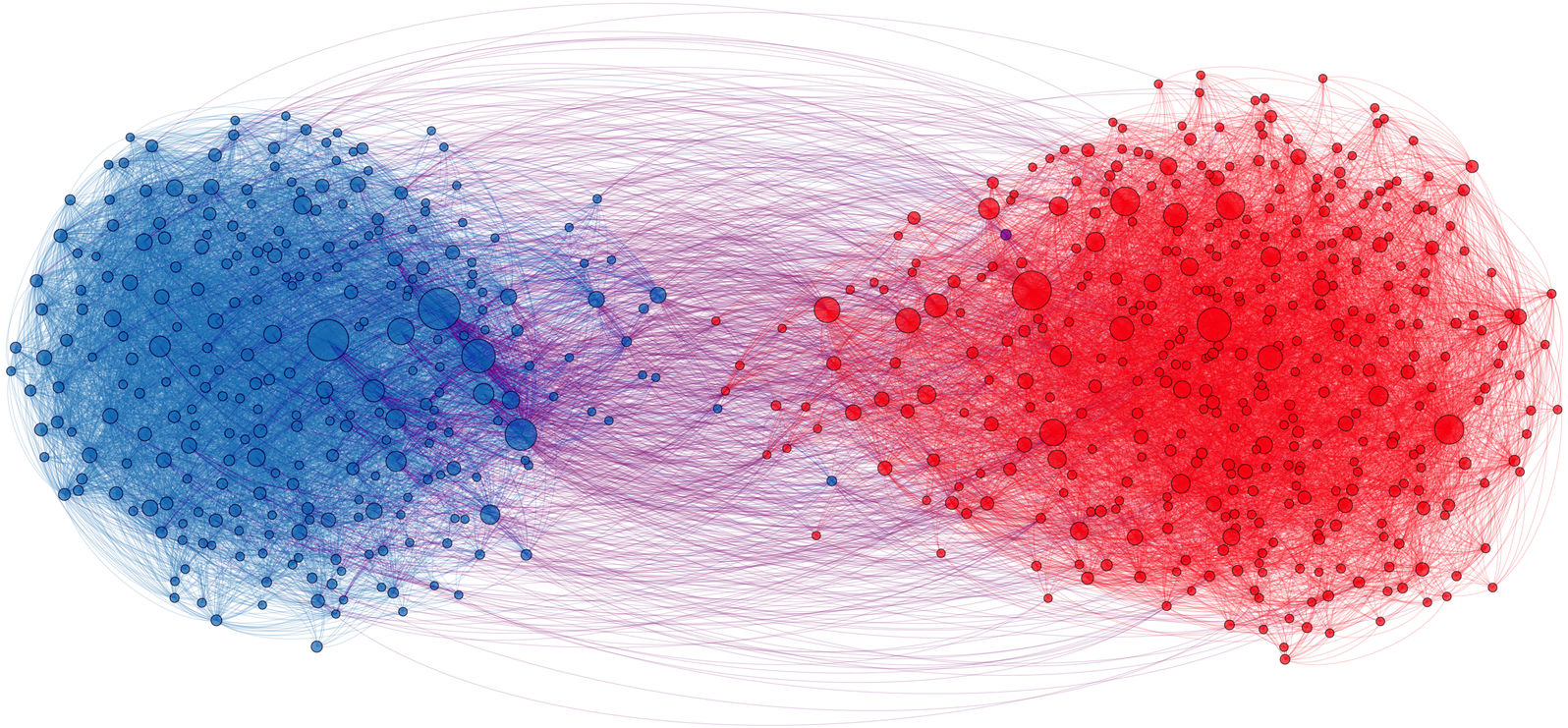


Social Networks in Human History

Online social networks have become predominant in last couple of decades. Billions of people are virtually connected.

Impacts and Effects of Social Networks

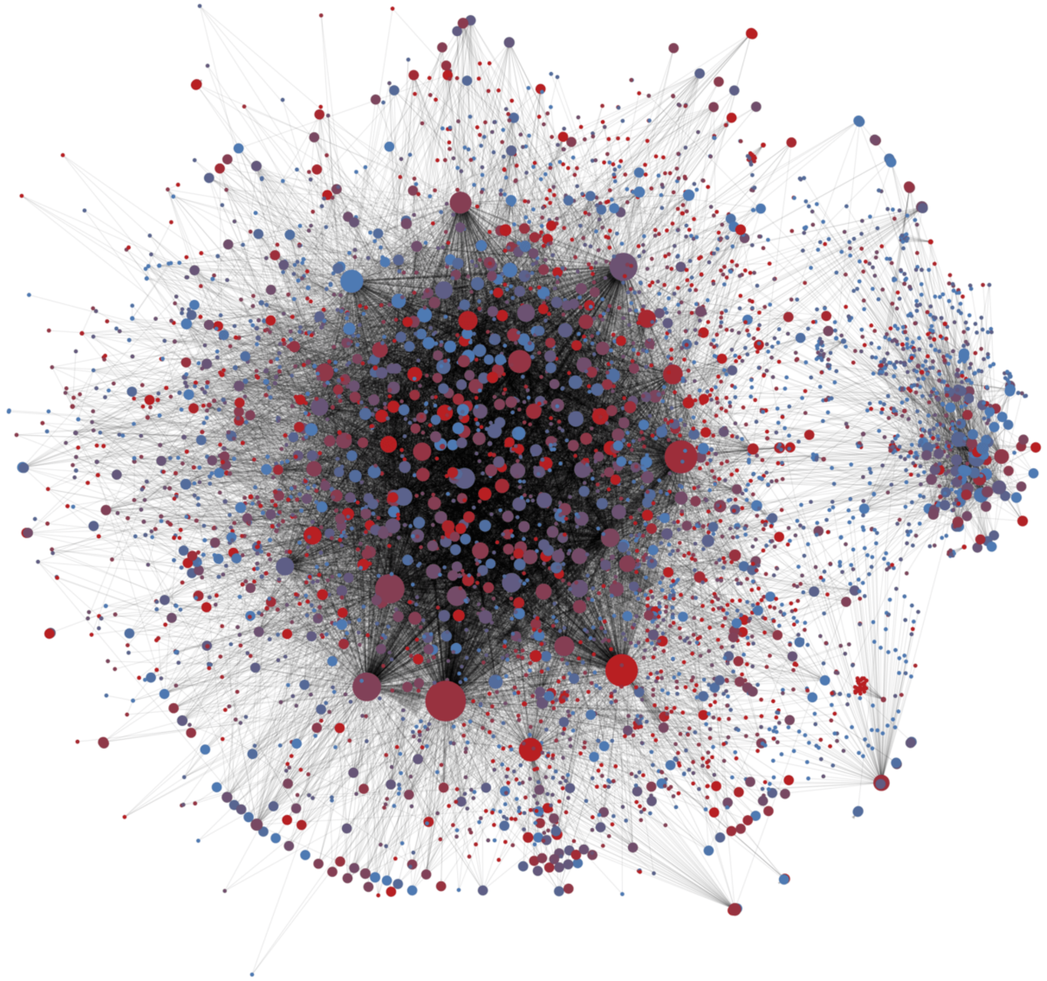
- Political polarization



Source: https://allthingsgraphed.com/

Impacts and Effects of Social Networks

- Political polarization
- (Mis)information spreading



Source: https://www.govtech.com/social/Misinformation-on-Social-Media-Can-Technology-Save-Us.html

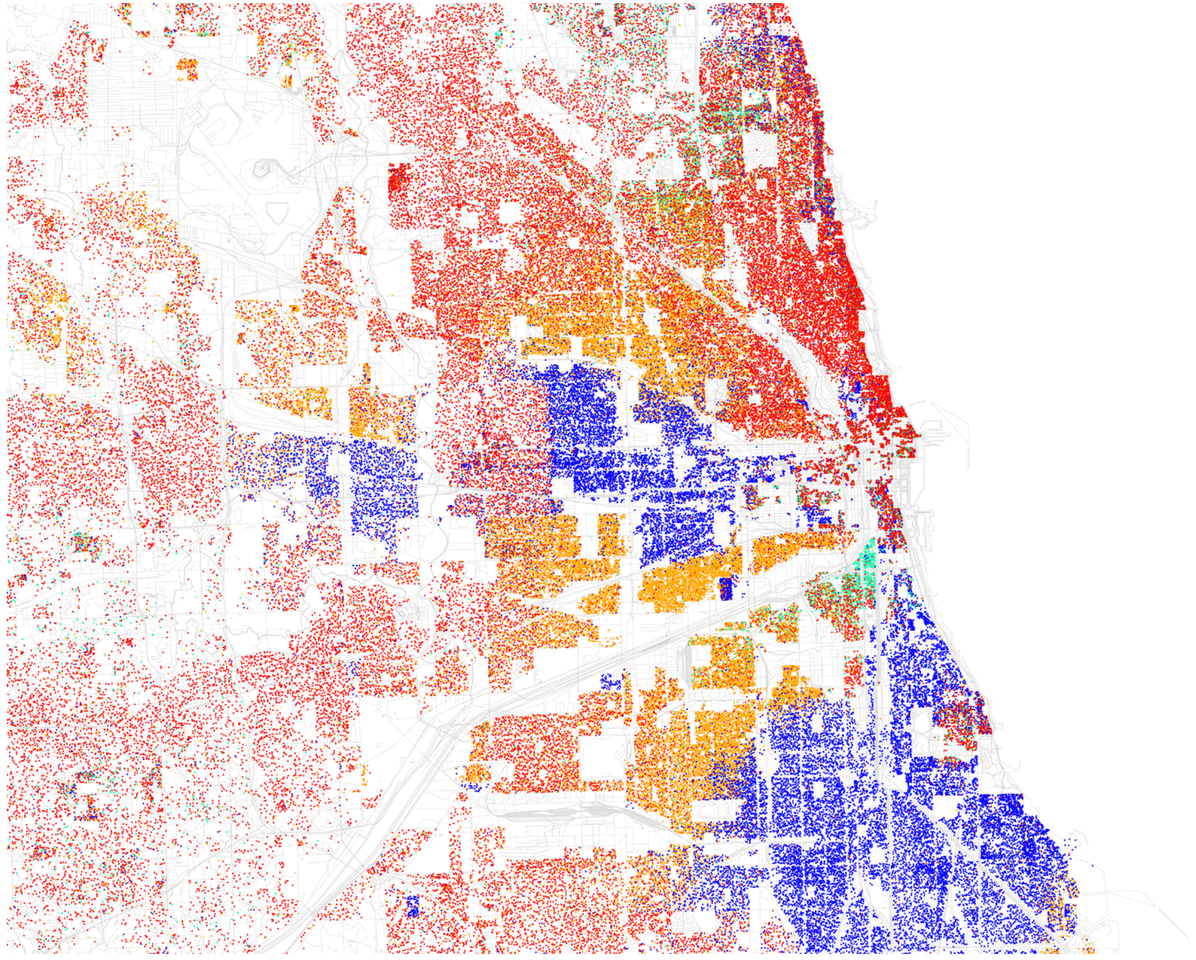
Impacts and Effects of Social Networks

- Political polarization
- (Mis)information spreading
- Disease spreading

Source: https://storymaps.arcgis.com/stories/4fdc0d03d3a34aa485de1fb0d2650ee0

Impacts and Effects of Social Networks

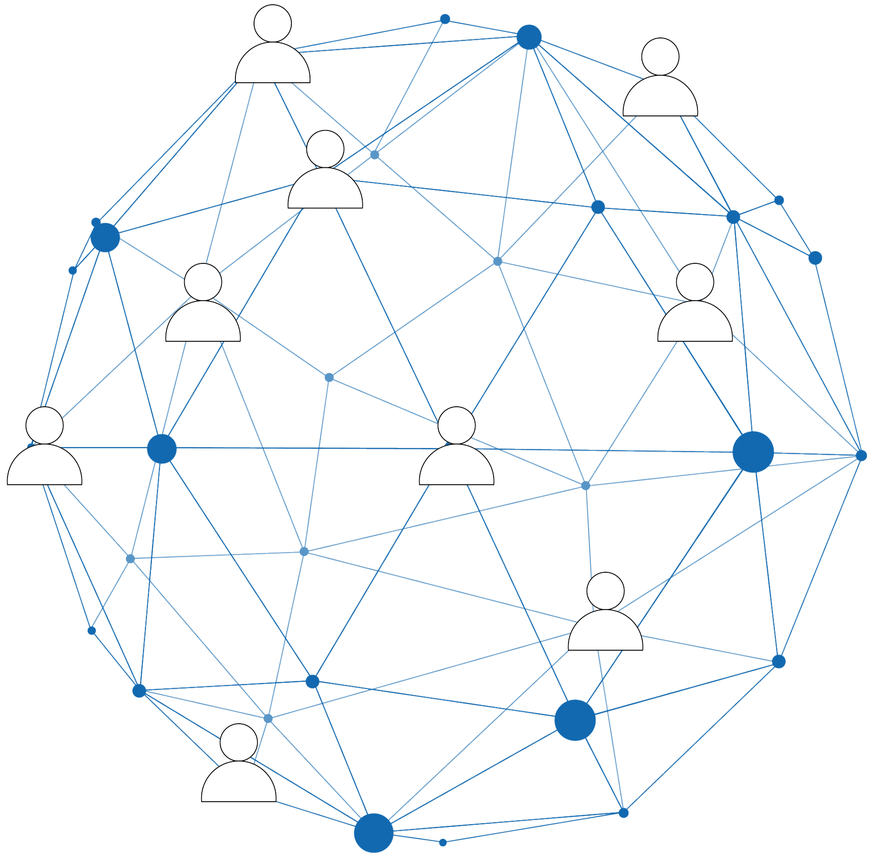
- Political polarization
- (Mis)information spreading
- Disease spreading
- Racial segregation
- Adoption of new technologies
- Diffusion of healthy behaviour, ...



Source: http://chicagostories.org/black-chicago/

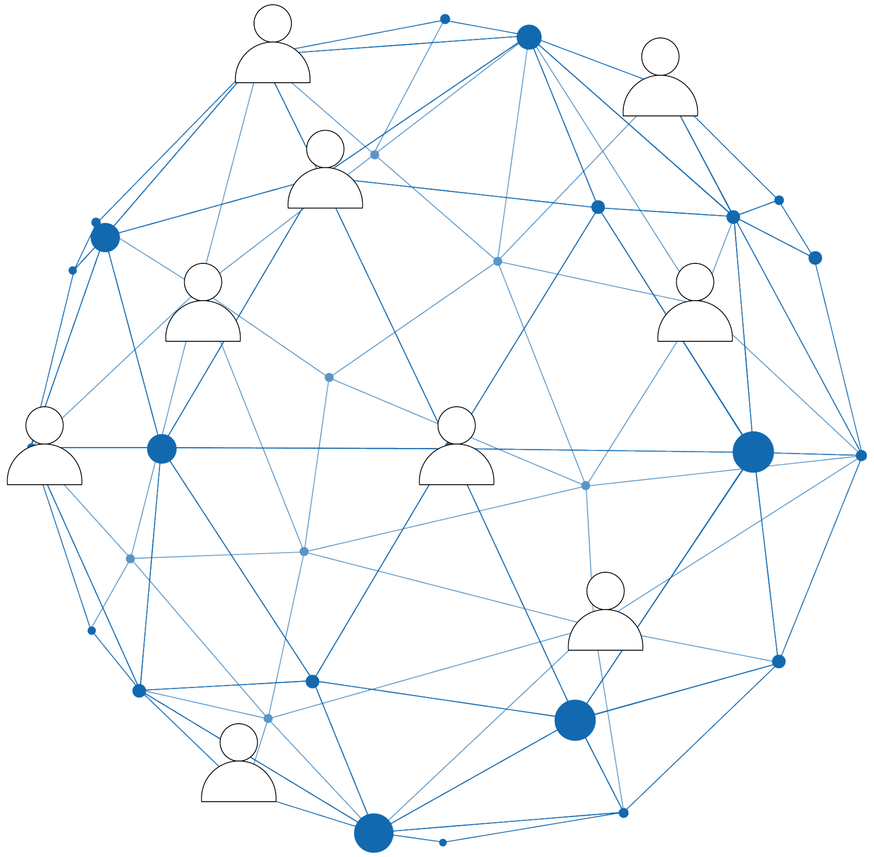
Objective

With the objective of advancing our understanding of these phenomena, we focus on the modelling, analysis, and inference of the social network formation process, which underlie many of them.



Preliminaries

- A social network is composed of nodes, representing actors/agents, and of ties or links, which correspond to their relationships.



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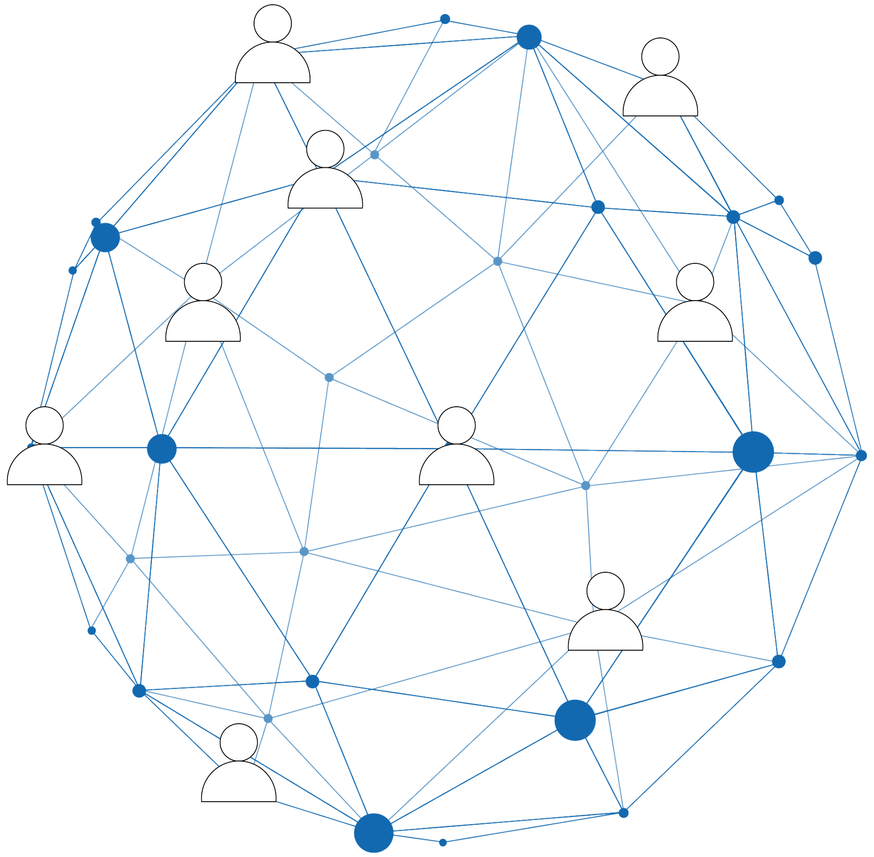

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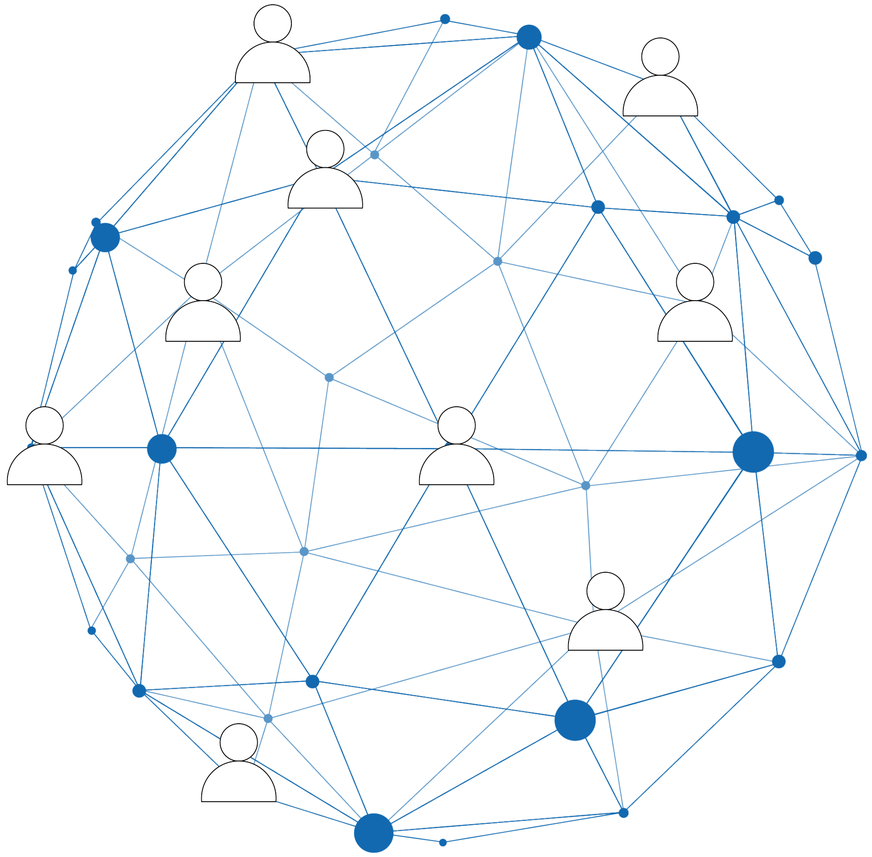
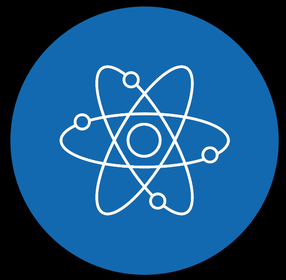
- Actors decide who they want to follow.
- Network positions provide benefits/costs to the actors.



Social Networks Complexity

Large-Scale Systems

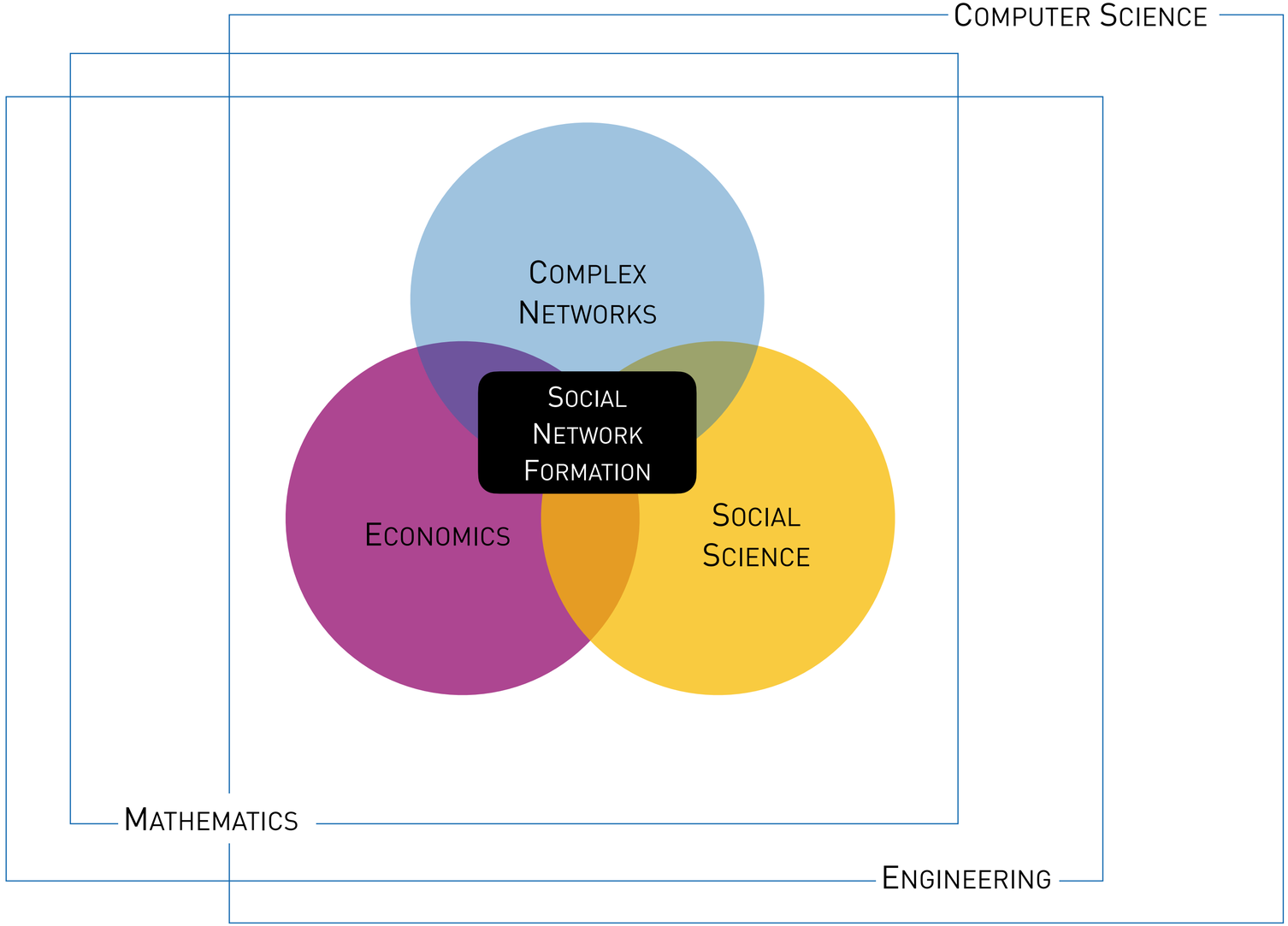
Human Behaviour



Interactions

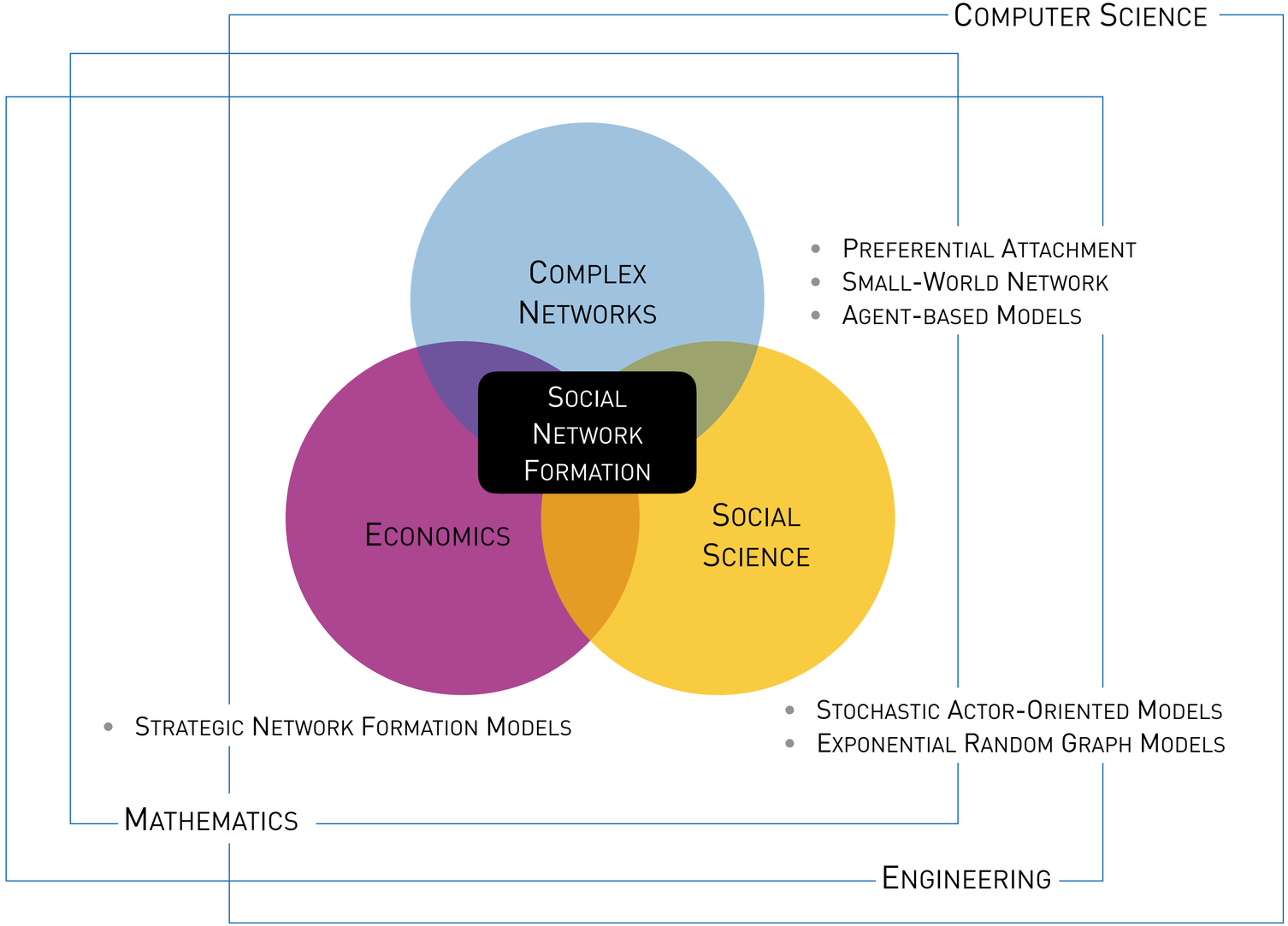
Social Network Formation Problem

Researchers from different communities study the problem of Social Network Formation.



Literature Review

Prior multidisciplinary research efforts on studying social network formation processes has privileged topological elements or socio-strategic incentives.



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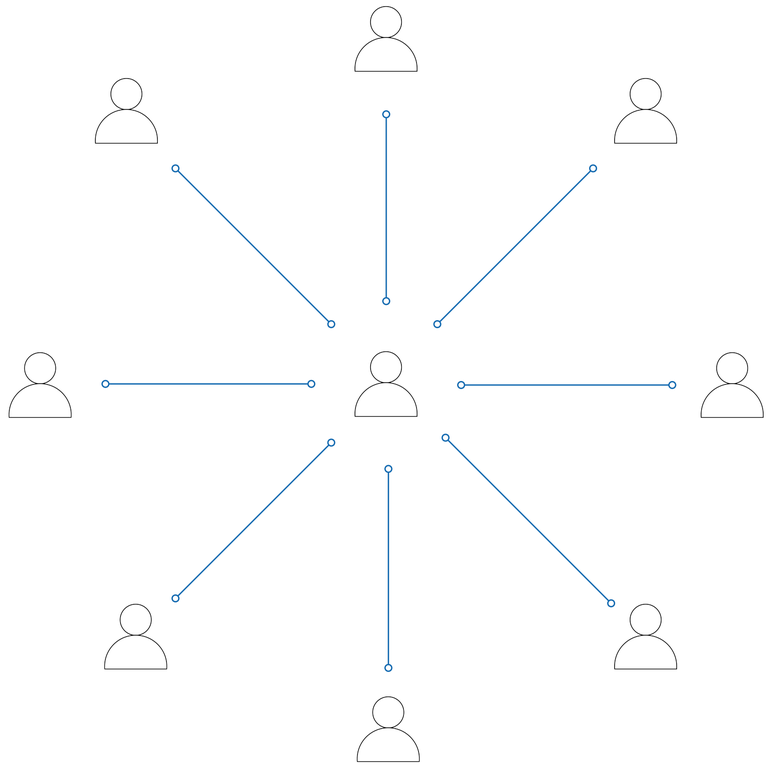

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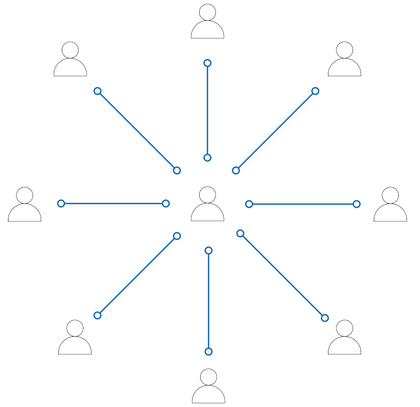

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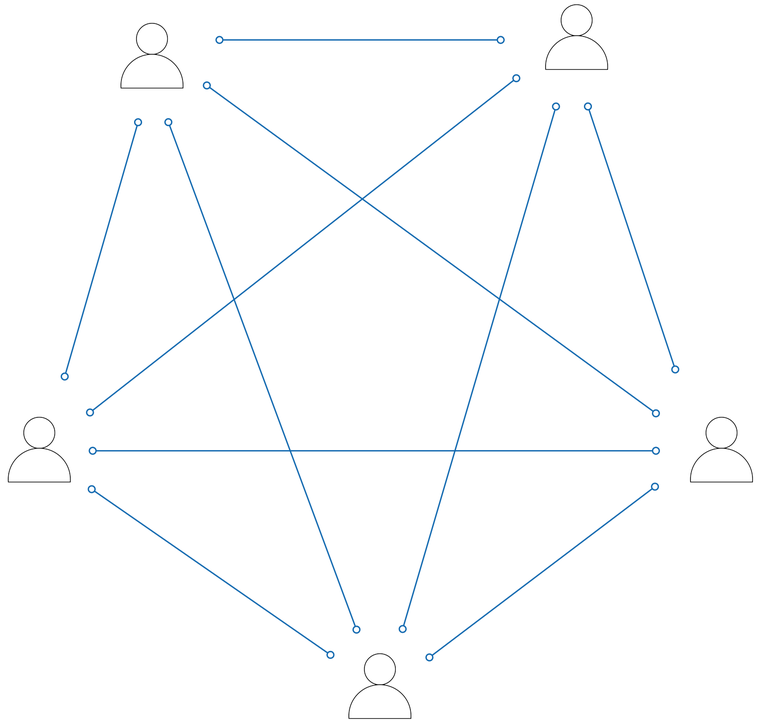

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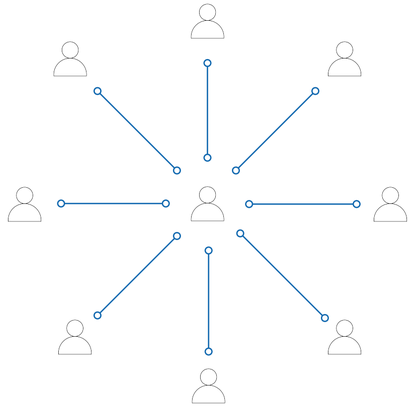
- Strategic Network formation models (e.g., eigenvector, betweenness, closeness, ...)
  Jackson, Matthew O. Social and economic networks. Princeton university press, 2010.



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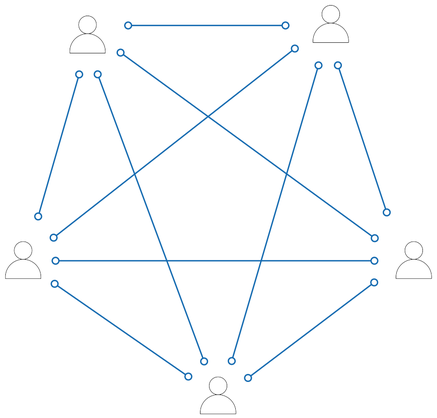

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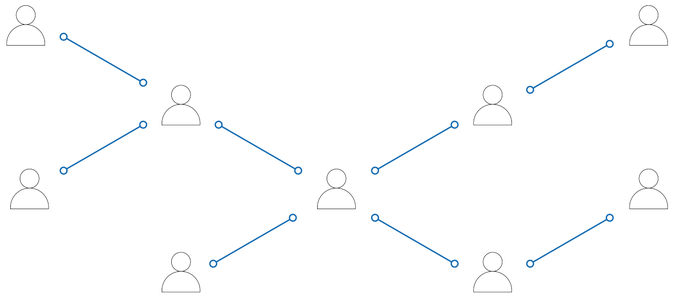

Betweenness Centrality

Offline vs Online Social Networks

Offline Social Networks

N. Pagan and F. Dörfler, “Game theoretical inference of human behaviour in social networks”, *Nature Communications, 2019*. 10.1038/s41467-019-13148-8

- Degree Centrality
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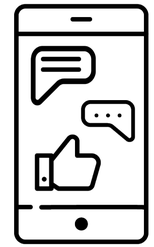

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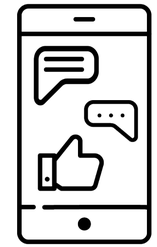


Online Social Networks

N. Pagan, W. Mei, C. Li, and F. Dörfler, "A meritocratic network formation model for the rise of social media influencers". Under review at *Nature Communications, 2021.* 10.21203/rs.3.rs-424956/v1

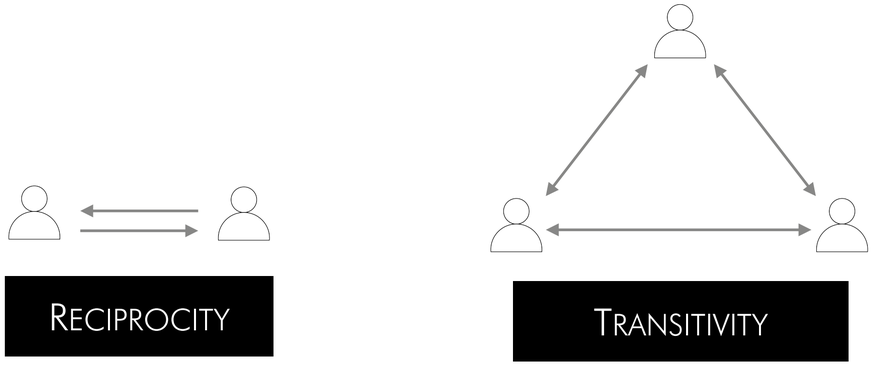
ONLINE SOCIAL NETWORKS:

QUALITY-BASED NETWORK FORMATION MODEL

Online Social Networks

- "Old-generation" online social networks try to mimic offline social networks.

Microscopic Sociological Principles



Reciprocity	100%
Transitivity	40%

[S. Teng, M. Yeh and K. Chuang, "Toward understanding the mobile social properties: An analysis on instagram photo-sharing network", 2015]

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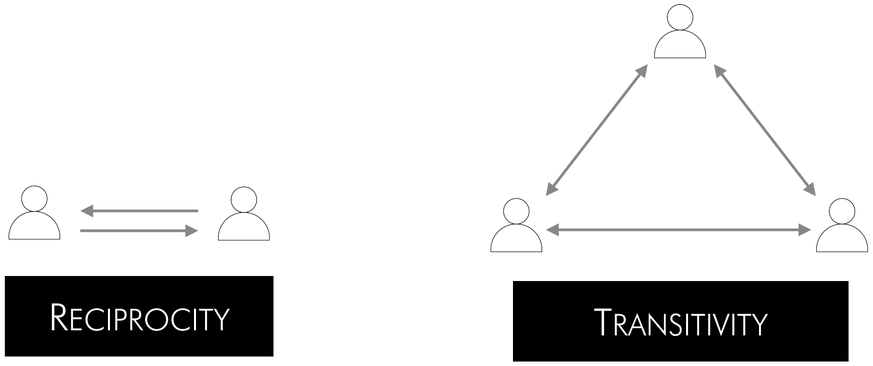

Reciprocity	100%	22%	14%
Transitivity	40%	23%	8%

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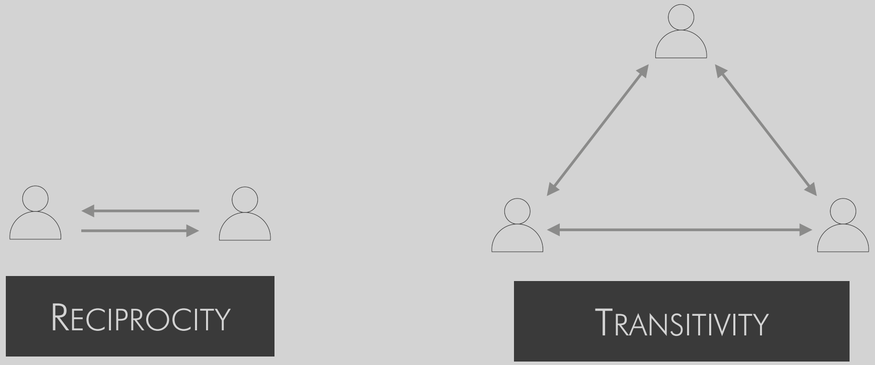

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- Users can contribute with their User-Generated Content (UGC): Tweets, Instagram photos, YouTube videos, Twitch streaming, ...

User-Generated Content

- Through the integrated search engines, users can search for their preferred content.

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- Intuitively, users are attracted by content that fits their interest, and which is of "high quality".

- To support this intuition, we collect and analyze a longitudinal Twitter data-set composed of social network scientists.

Meritocratic Principle

1. We enumerate the agents in decreasing-quality order (agent 1 is the highest quality, agent 2 is the second highest quality, ...).

Rank

- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8

Meritocratic Principle

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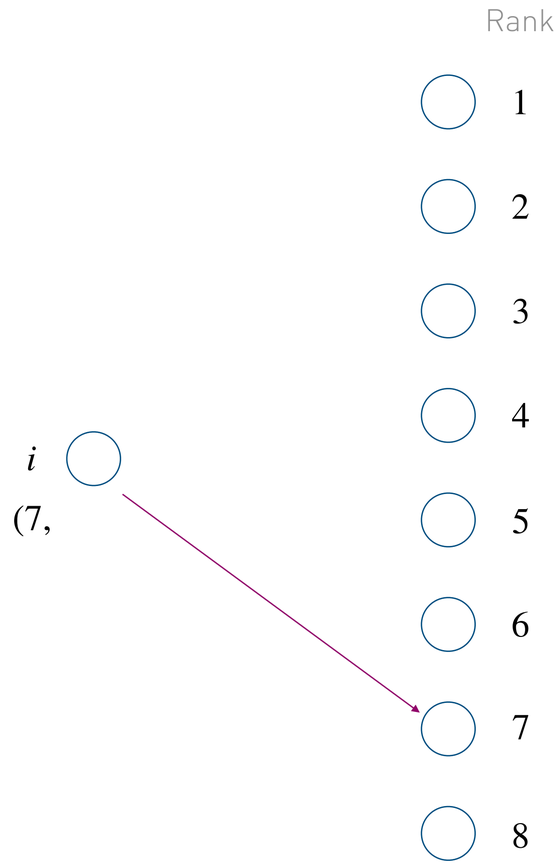
2. For each agent i, let $\left(i_1, i_2, \ldots, i_{d_i^{out}}\right)$ be the temporal sequence of outgoing connections such that i_j is the rank of the node that i started following at the j^{th} time-step.



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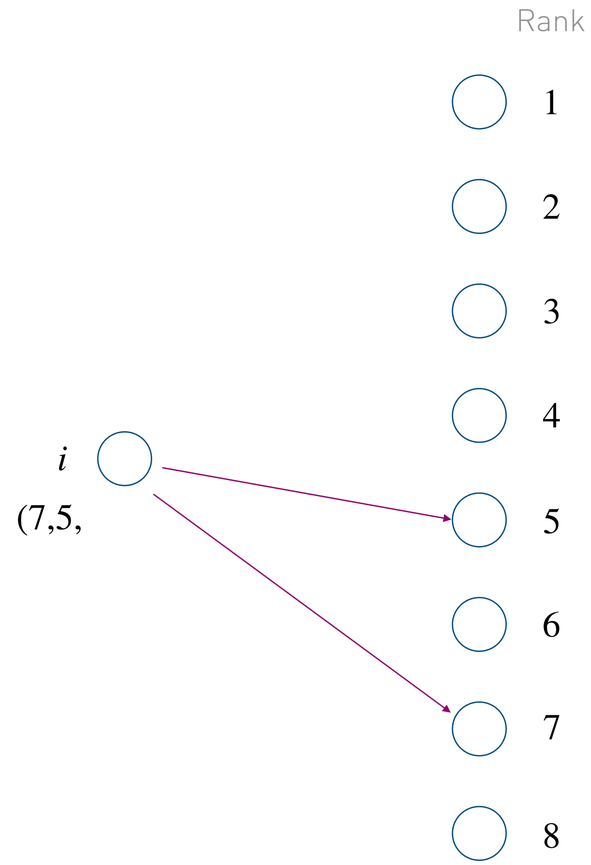

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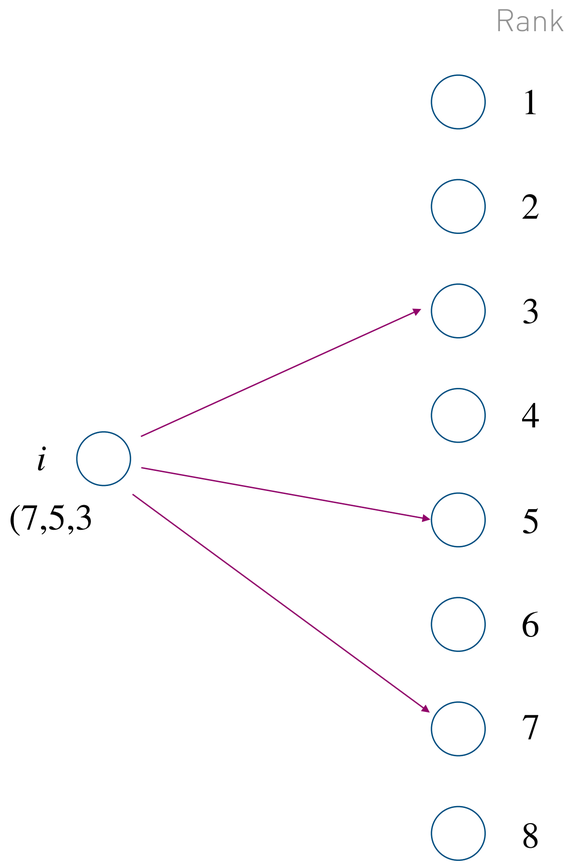

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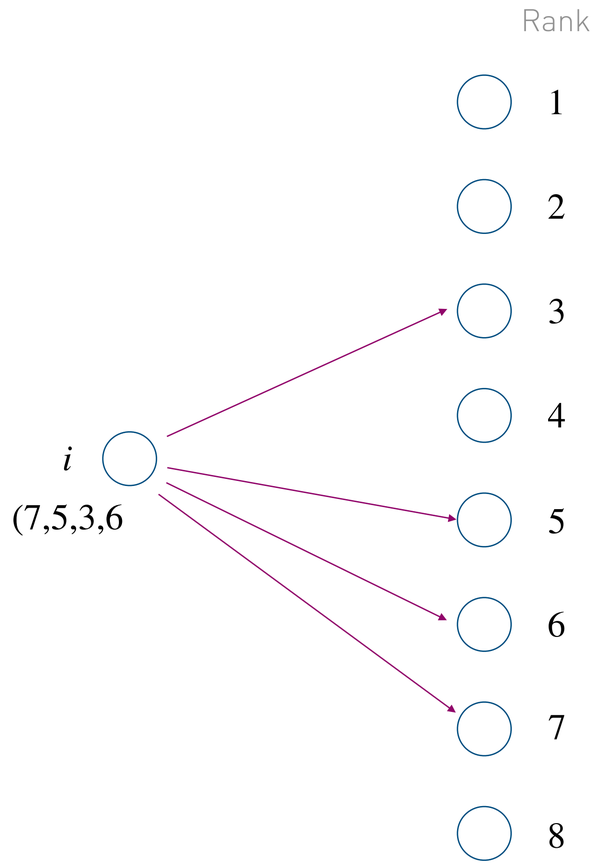

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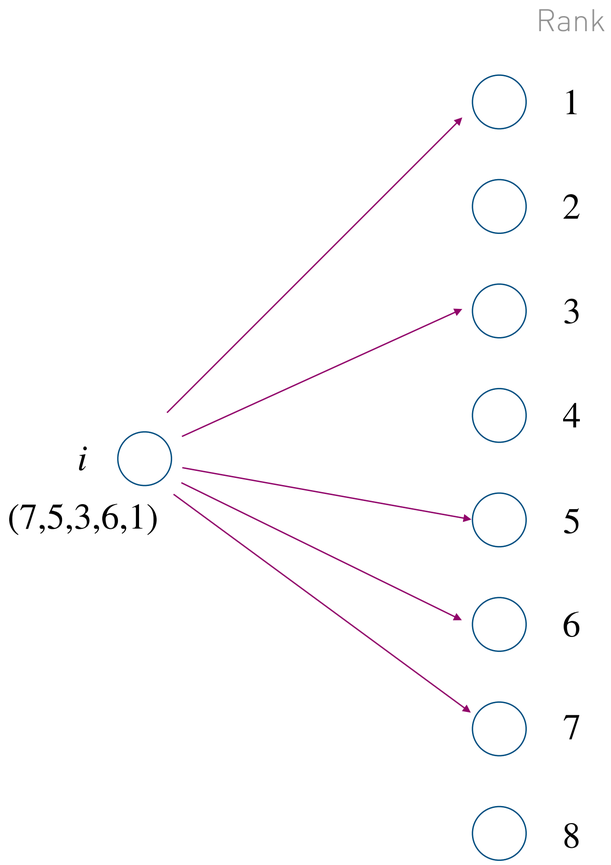

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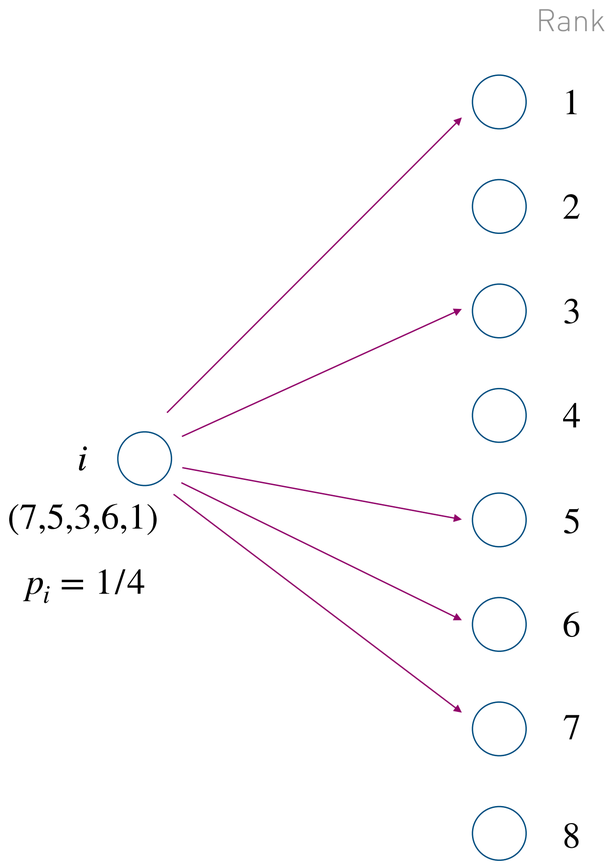
3. Then,

$$p_i = \frac{\left| \left\{ j \in \{2,\ldots,d_i^{out}\}, \text{ s.t. } i_j > \text{median} \left\{ i_1, \ldots, i_{j-1} \right\} \right\} \right|}{d_i^{out} - 1},$$

is the probability that a new connection of node i has lower quality than the median of the previous connections.



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$p_i = 0.5$ means complete random process, while $p_i < 0.5$ means agents tend to follow higher quality nodes compared to those already followed.

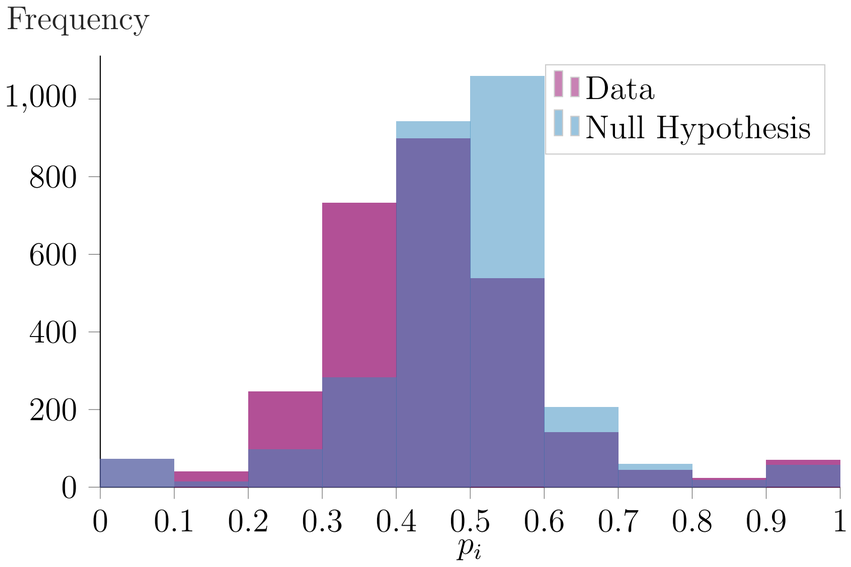

Fig. We compare the ordered temporal sequence (purple) of outgoing connections in a Twitter data-set on a network of social network scientists with a random temporal sequence (blue). The difference between the distributions is statistical significant.

MODEL

Quality-Based Network Formation Model

- Every agent i is attributed with a quality parameter $q_i \in [0,1]$ describing the likelihood her content will be liked by others. Wlog, $q_1 > q_2 > \ldots > q_N$.

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$$a_{ij}(t+1) = \begin{cases} 1, & \text{if } q_j > V_i(t) := \max_{j \in \mathcal{F}^{i,\text{in}}(t)} q_j, \\ a_{ij}(t), & \text{otherwise,} \end{cases}$$

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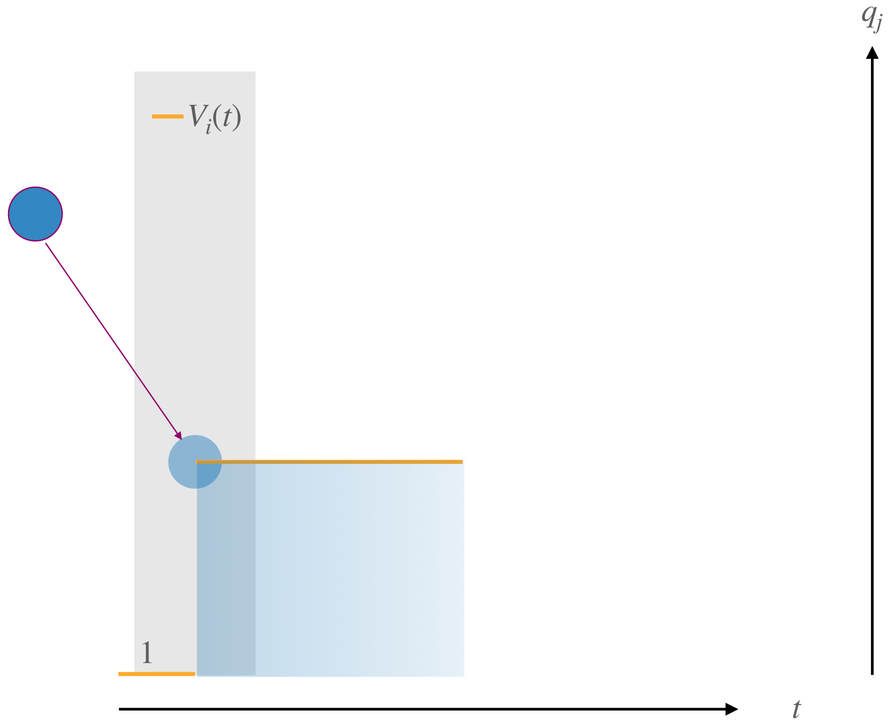

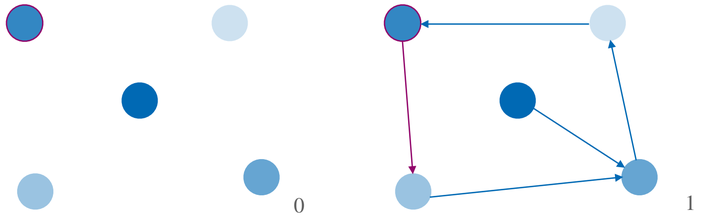

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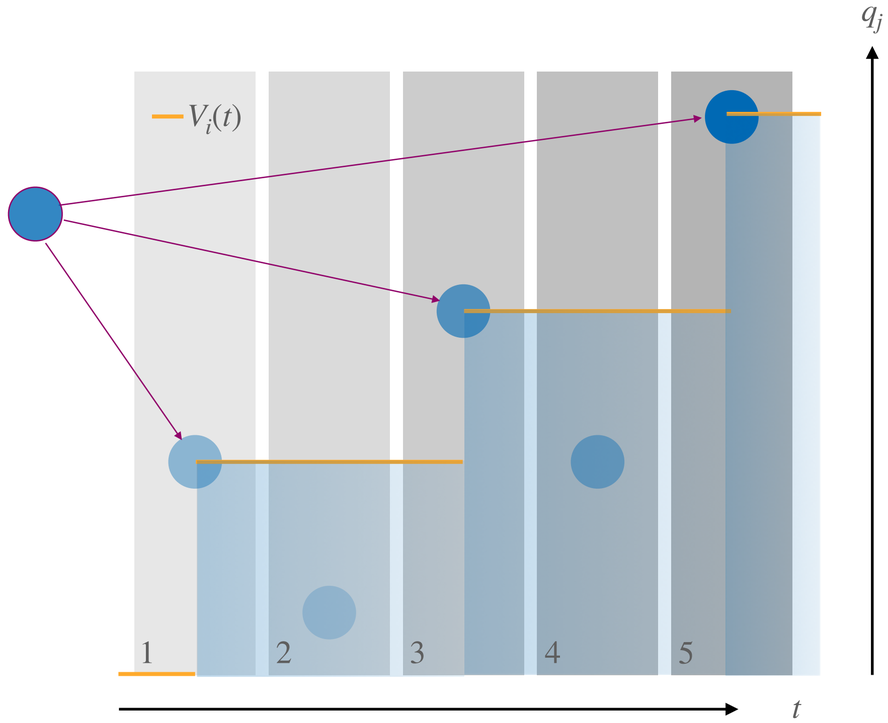
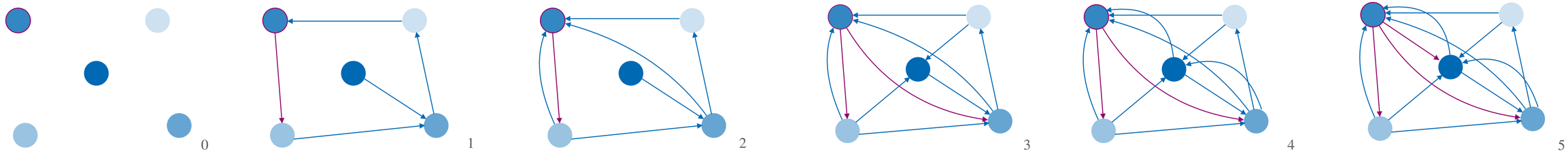
- Every agent i is attributed with a quality parameter $q_i \in [0,1]$ describing the likelihood her content will be liked by others. Wlog, $q_1 > q_2 > \ldots > q_N$.

- Each agent i sequentially meets another distinct agent j (e.g., uniformly at random) and decides whether to start following j or not according to:

$$a_{ij}(t+1) = \begin{cases} 1, & \text{if } q_j > V_i(t) := \max_{j \in \mathcal{F}^{i,\text{in}}(t)} q_j, \\ a_{ij}(t), & \text{otherwise,} \end{cases}$$

where $\mathcal{F}^{i,\text{in}}(t) = \left\{ j, \text{ s.t. } a_{ij}(t) = 1 \right\}$ denotes the set of i's followees.





RESULTS

Theoretical Analysis

1. Convergence. The dynamical process reaches an equilibrium state almost surely.

Theorem (Convergence).

For any set of qualities $\{q_1, \ldots, q_N\}$, the network reaches an equilibrium almost surely.

Theoretical Analysis

1. Convergence. The dynamical process reaches an equilibrium state almost surely.

2. Indegree distribution. The meritocratic principle leads to a strong correlation between quality and indegree.

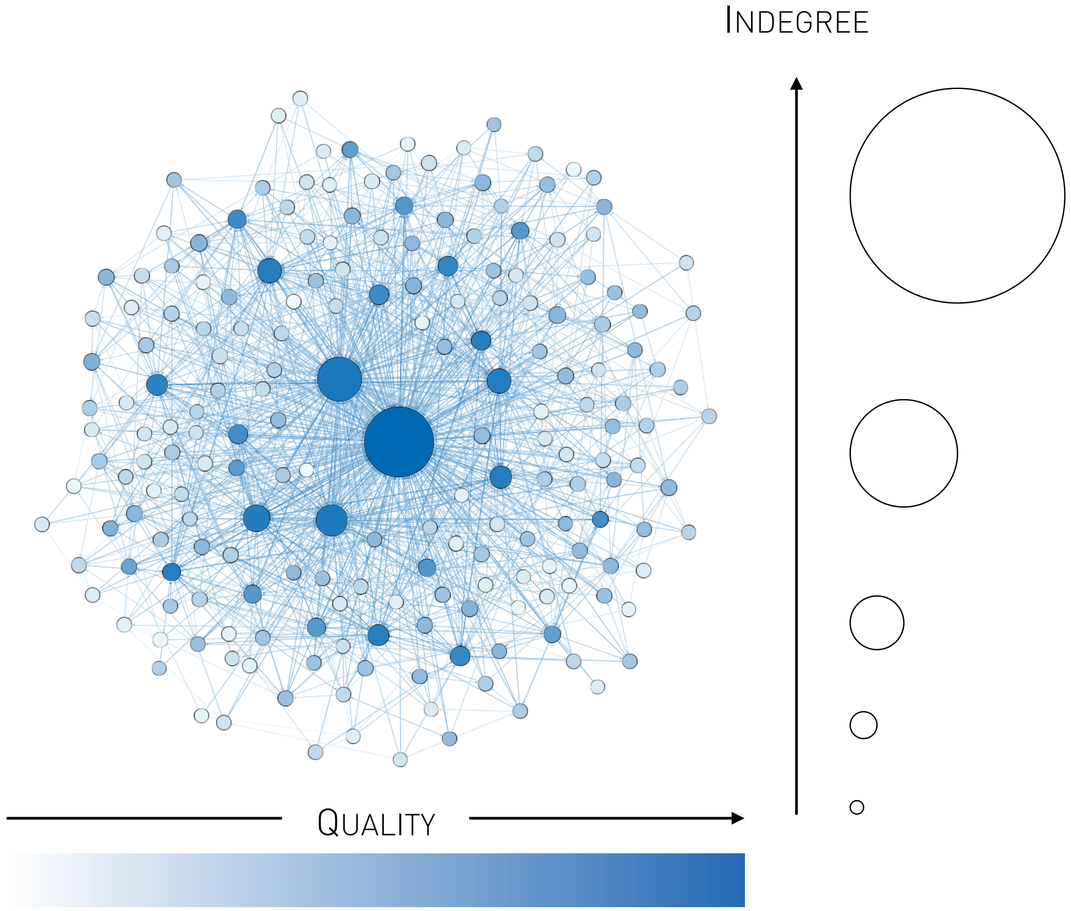


Fig. Numerical simulation results of a network of N=200 agents until equilibrium.

Theoretical Analysis

1. Convergence. The dynamical process reaches an equilibrium state almost surely.

2. Indegree distribution. The meritocratic principle leads to a strong correlation between quality and indegree, which is named Zipf's law.

Theorem (Indegree Distribution).

At equilibrium, the expected indegree follows a Zipf's law:

$$\mathbb{E}(d_i^{in,\star}) = \begin{cases} N-1, & \text{if } i = 1, \\ \frac{N}{i}, & \text{otherwise.} \end{cases}$$

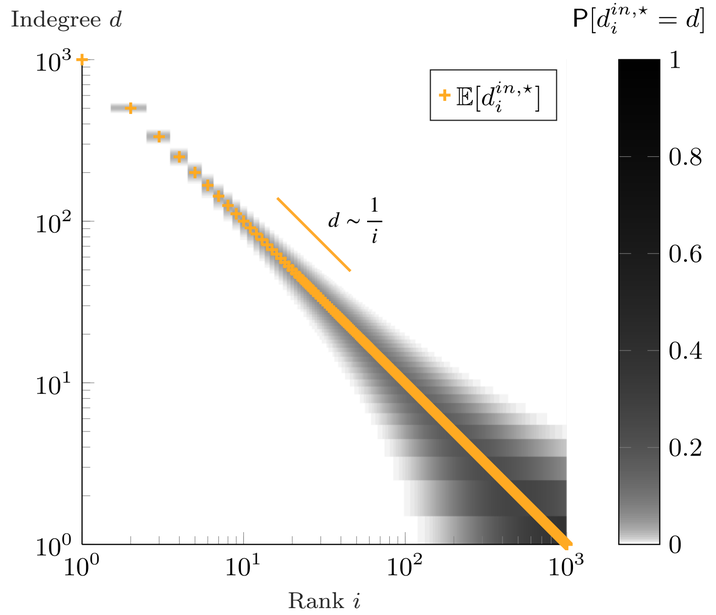


Fig. Theoretical probability density function of the nodes' indegree in a network of N=1000 agents.

Theoretical Analysis

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$$\mathsf{P}[d^{in,\star} = d] = \frac{1}{N}\sum_{i=1}^{N} \mathsf{P}[d_i^{in,\star} = d]$$

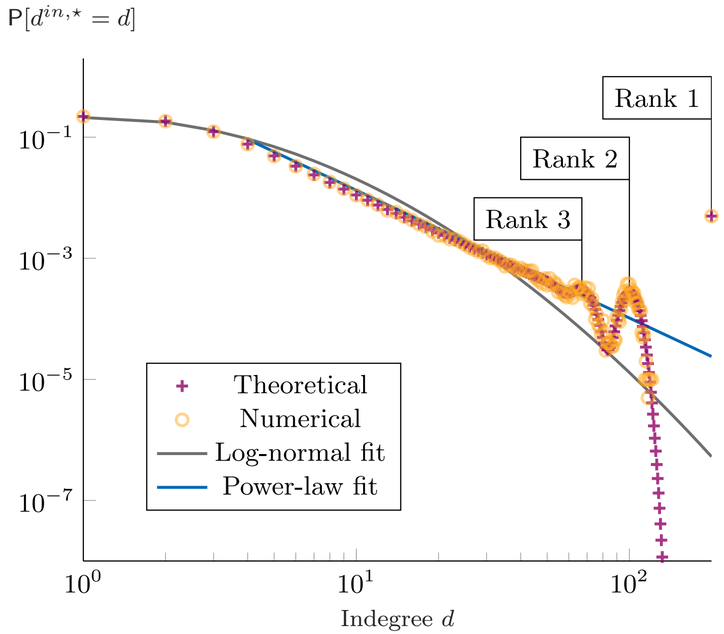

Fig. Theoretical probability density function of the average nodes' indegree in a network of N=1000 agents.

Theoretical Analysis

1. Convergence. The dynamical process reaches an equilibrium state almost surely.

2. Indegree distribution. The meritocratic principle leads to a strong correlation between quality and indegree, which is named Zipf's law.

3. Outdegree distribution. The nodes' outdegree remains bounded because the agents quickly find top quality nodes.

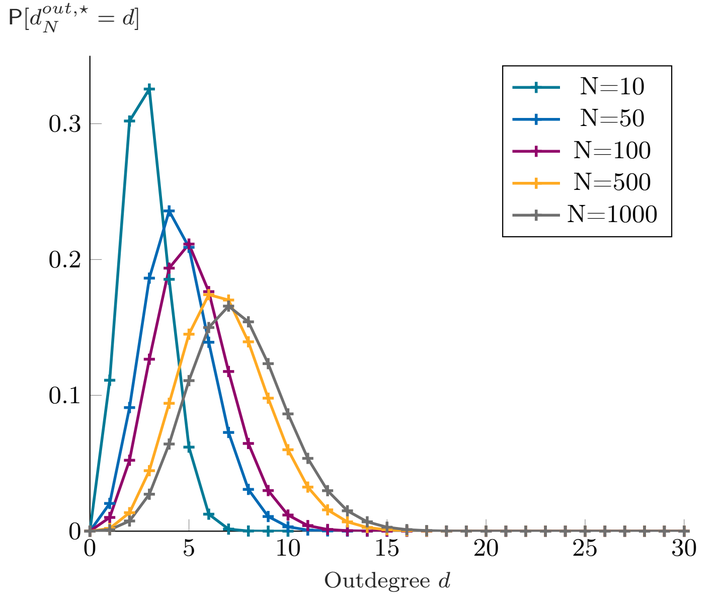

Fig. Theoretical outdegree distribution function for different network sizes.

Theoretical Analysis

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Theorem (Outdegree Distribution).

The nodes' expected outdegree $\mathbb{E}\left(d_N^{\text{out},\star}\right)$ in a network of $N \geq 2$ agents equals the $(N-1)$-th harmonic number:

$$\mathbb{E}\left(d_N^{\text{out},\star}\right) = \sum_{k=1}^{N-1} \frac{1}{k}.$$

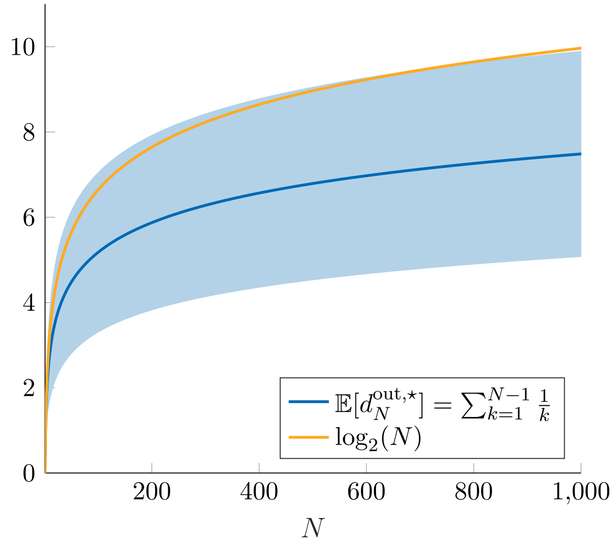

Fig. Theoretical expected outdegree as a function of the network size.

Theoretical Analysis

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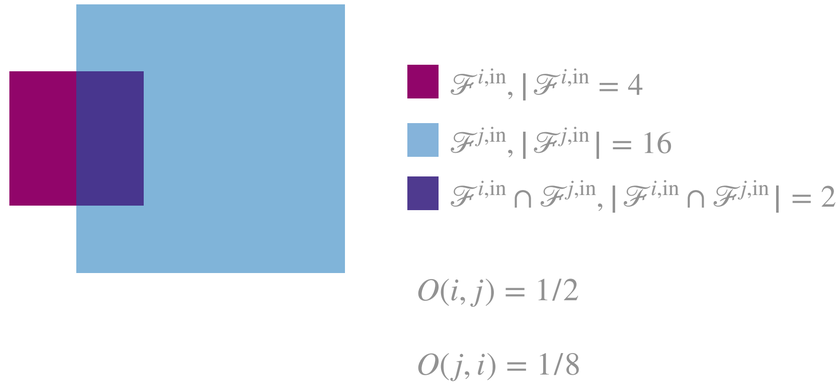
4. Audience overlap. The overlap among the followers' sets is significantly high, with a regular pattern.

Definition (Audience Overlap).

We define the Audience Overlap index as follows:

$$O(i,j) := \frac{\left|\mathcal{F}^{i,\text{in}} \cap \mathcal{F}^{j,\text{in}}\right|}{\left|\mathcal{F}^{i,\text{in}}\right|} \in [0,1], \quad \text{if } \left|\mathcal{F}^{i,\text{in}}\right| > 0,$$

and 0 otherwise, where $\mathcal{F}^{i,\text{in}}$ denotes the set of followers of i.



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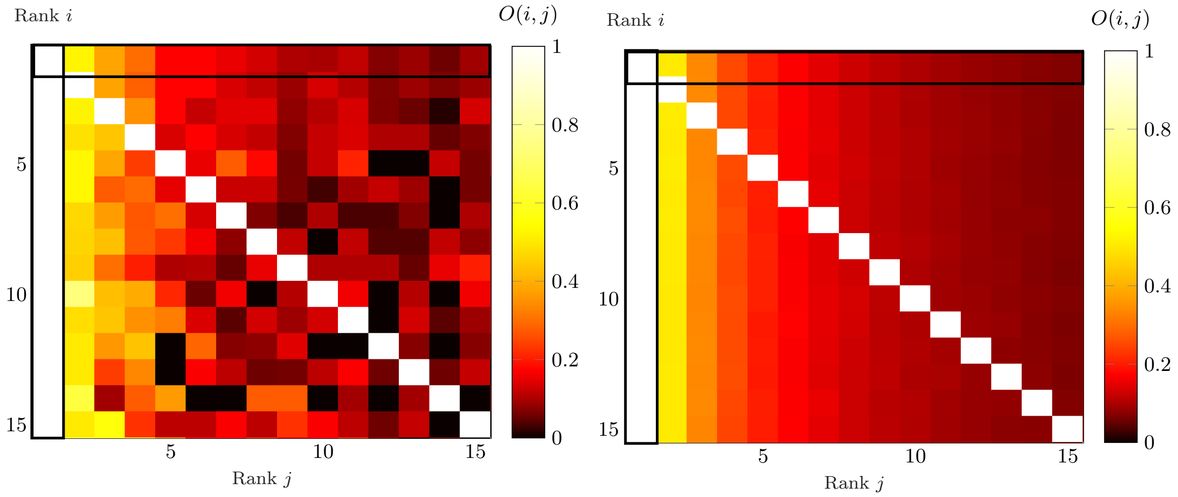

Fig. Numerical results on the followers' overlap among the top 15 nodes of a network of 1000 agents. Left: one simulation. Right: average across 1000 simulations.

Real-World data comparison

To validate our model:

1. We selected several different categories from Twitch, e.g., chess or poker.

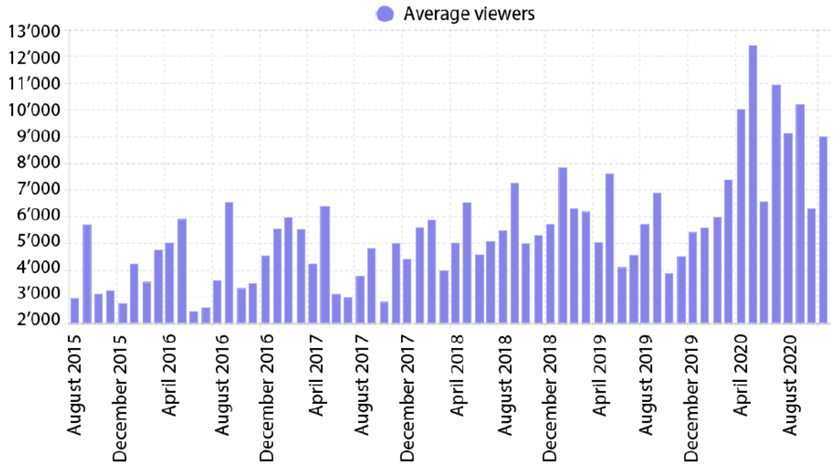

Fig. Average viewers in the poker category of Twitch.

Real-World data comparison

To validate our model:

1. We selected several different categories from Twitch, e.g., chess or poker.
2. For each of these categories, we reconstructed the underlying bipartite networks of streamers and followers.

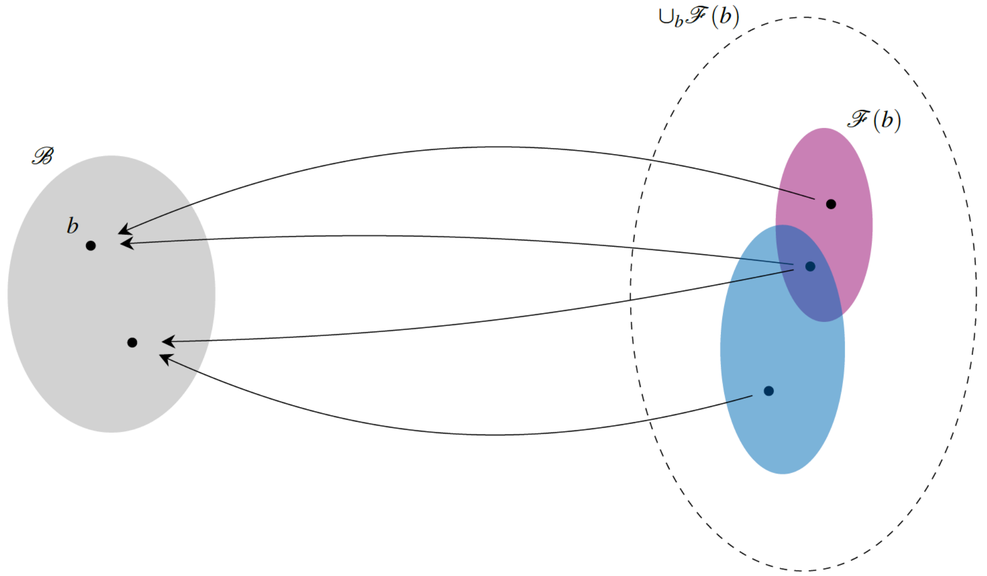

Fig. Sketch of the bipartite network composed of the set of broadcasters (on the left) and their followers (on the right).

Real-World data comparison

To validate our model:

1. We selected several different categories from Twitch, e.g., chess or poker.
2. For each of these categories, we reconstructed the underlying bipartite networks of streamers and followers.
3. We analyzed the network in terms of indegree and outdegree distribution, as well as of the audience overlap index.

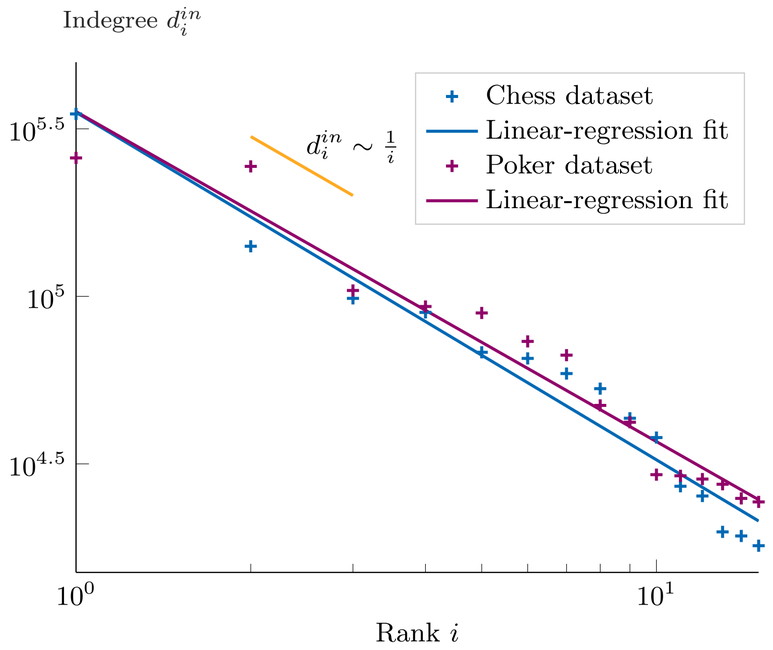

Fig. Empirical results of the indegree-rank distribution are aligned with a Zipf's law (regression coefficient ~ -1).

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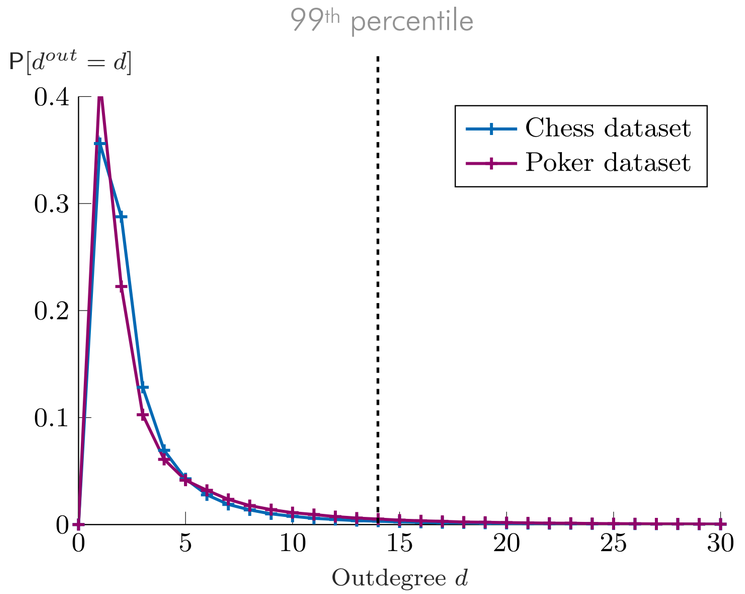

Fig. The empirical outdegree distribution shows some discrepancies which are caused by a larger number of low outdegree nodes. This phenomenon requires further investigation.

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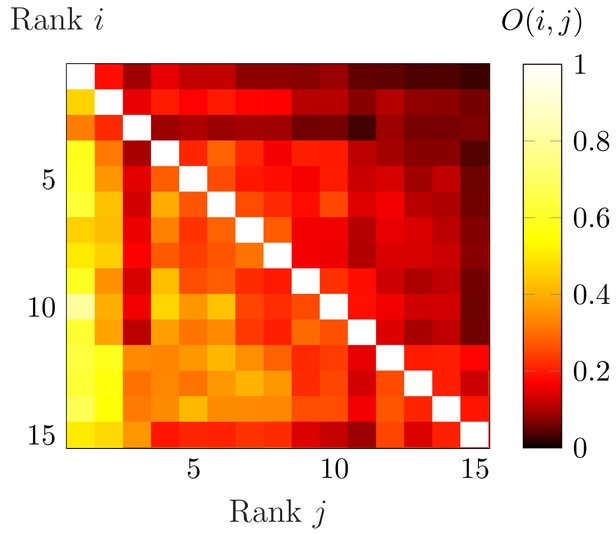

Fig. The empirical audience overlap index shows the expected decreasing trend (from left to right) and the homogeneity across the rows.

CONCLUSIONS

Summary

1. Inference. We collect and analyze a longitudinal Twitter data-set that allows to infer a meritocratic principle.

2. Model. We propose network formation process based on the quality of the UGC and on the meritocratic principle.

3. Analysis. We analysed the properties of the resulting network in terms of indegree and outdegree distributions, as well as with a newly defined audience overlap index.

4. Validation. We validate our estimation results against data-sets collected on Twitch.

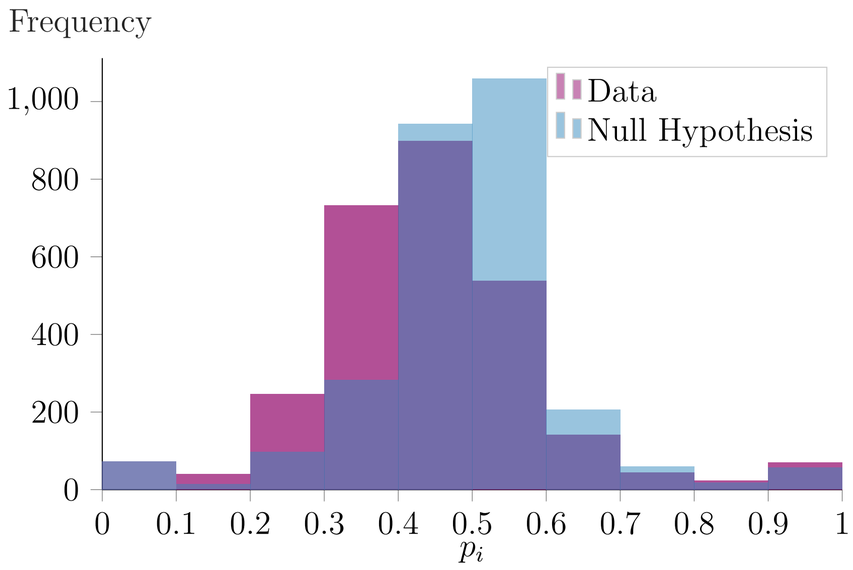

Fig. We compare the ordered temporal sequence (purple) of outgoing connections in a Twitter data-set on a network of social network scientists with a random temporal sequence (blue). The difference between the distributions is statistical significant.

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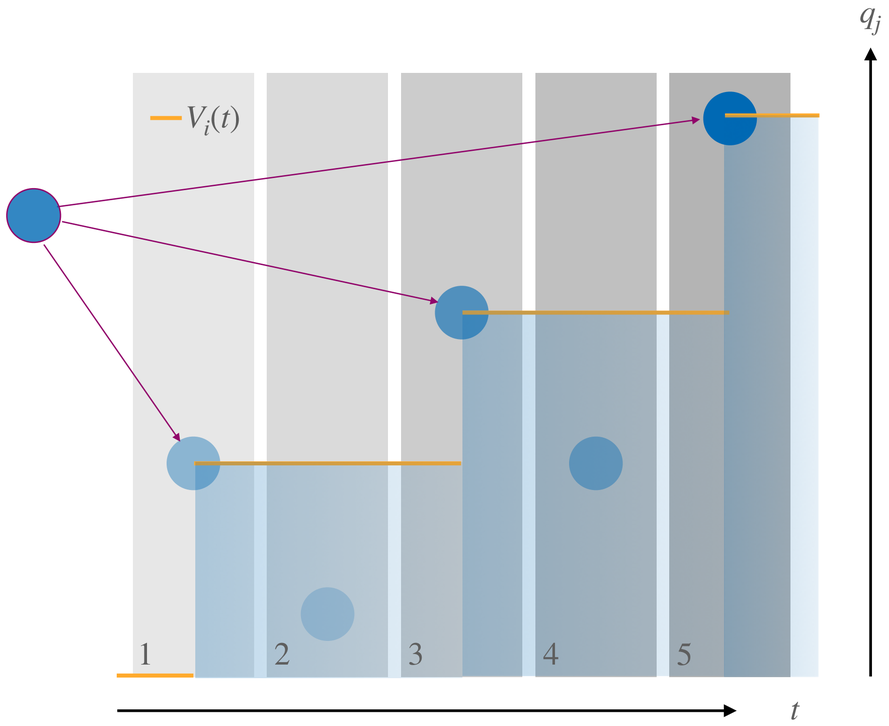

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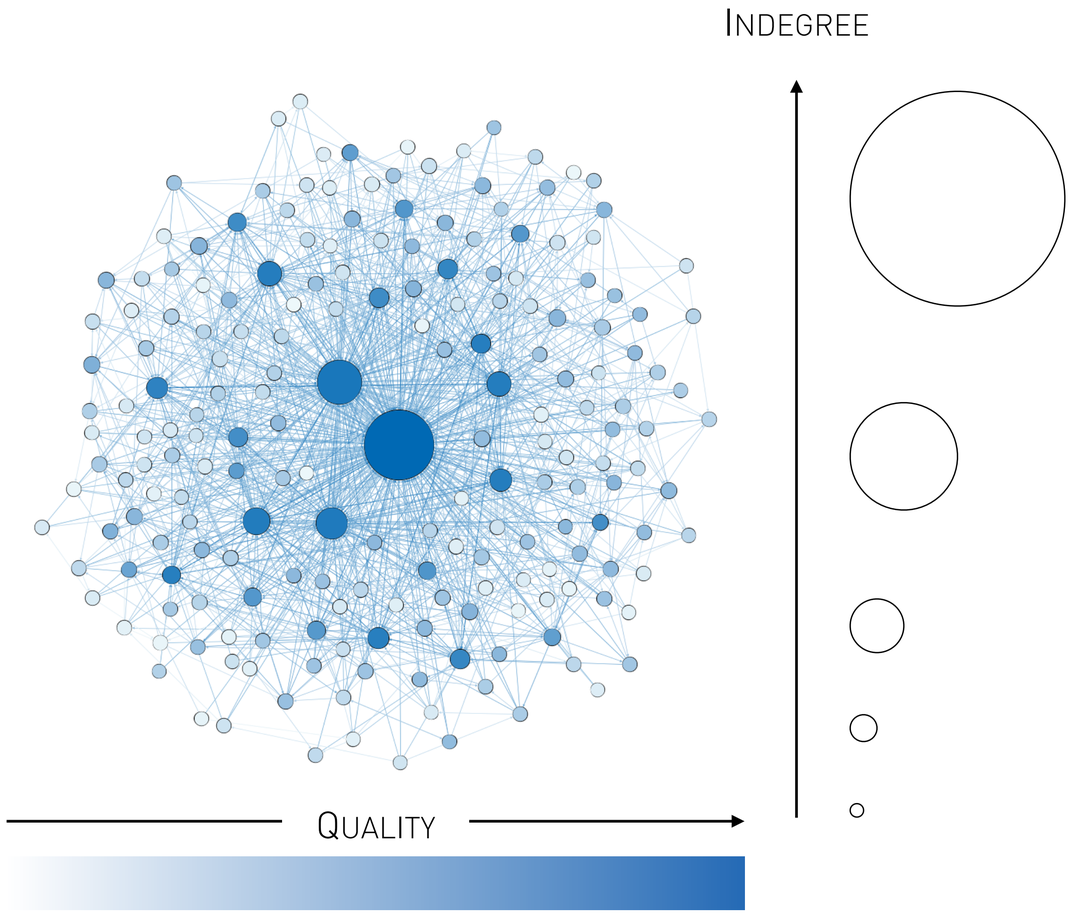

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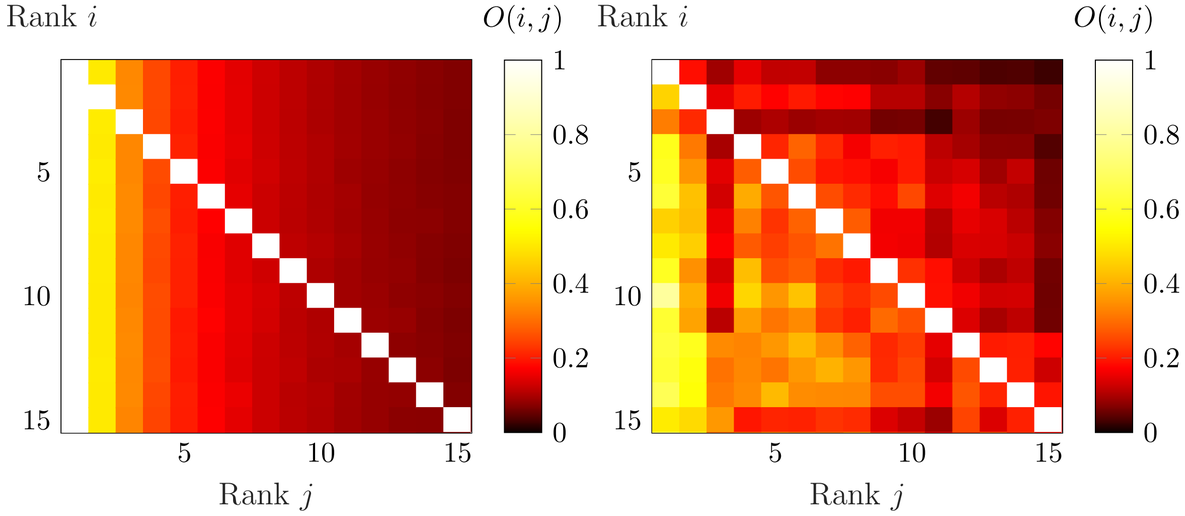

Fig. Left: simulation results. Right: empirical data analysis from Twitch data-set.

Outlook

- Our objective was to propose a new social network formation model rooted on the User-Generated Content and on its quality.
- Despite its simplicity, the model and its meritocratic principle are in agreement with empirical data from Twitch.

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- Despite its simplicity, the model and its meritocratic principle are in agreement with empirical data from Twitch.

- Further investigate and validate the meritocratic principle acting on (other) UGC-based online social networks.
- Analyze the properties of the resulting networks in terms of spreading of information, or mis-information.
- Study the effect of the recommendation systems and the interplay between humans and social network algorithms.

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Prof. Dr. Florian Dörfler

Prof. Dr. Wenjun Mei

Cheng Li

Nicolò Pagan

nicolo.pagan@uzh.ch - pagann@ethz.ch

people.ee.ethz.ch/~pagann/

University of Zürich - Social Computing Group

ETH Zürich - Social Networks Lab

BACK UP SLIDES